Final Proposal

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E-Cigarette Health Study

Research Question

Are those with healthier lifestyles less likely to use e-cigarettes than those who may have less healthy lifestyles?

Research Motivation

There has been a lot of discussion in modern society over the use of e-cigarettes. Are they less harmful than traditional cigarettes or do they actually cause just as much damage themselves? There have been studies that have shown that e-cigarette usage may be less harmful than traditional cigarettes since they do not contain as many dangerous ingredients (Eaton et al.; 2018). However, they do contain a lot of nicotine and may actually be more addictive than traditional cigarettes. An e-cigarette gives its users a nicotine-containing aerosol by heating up a solution that contains the nicotine, the flavoring agent, and other ingredients (Grana et al.; 2014). The long-terms effects of e-cigarettes are also not known due to their relatively short existence. There is, however, plenty of evidence that shows that e-cigarettes vary in terms of what they emit based on the flavoring of the e-cigarette and the device that is being used to operate them. So, who is more likely to use an e-cigarette? Are those with

a healthier lifestyle more likely to use it since they are supposedly less harmful? Or, are these individuals less likely due to their already healthier lifestyle choices?

Niche

This research will allow for a prediction for whether an individual is more likely to use an e-cigarette. This could be useful to parents and students involved in a Drug Awareness campaign along the lines of D.A.R.E. (Drug Abuse Resistance Education). D.A.R.E.'s main purpose is to educate students and parents about the dangers of using drugs and alcohol (D.A.R.E., 2024). D.A.R.E. could provide lessons where they look at the lifestyle choices that an individual has and see if they are more likely to use an e-cigarette. They could then use that information to try to help them to make a better decision to be healthier. Parents want their children to make healthier decisions, so this could be useful to them. These parents could look for lifestyle choices that may mean that their children have a higher likelihood of using e-cigarettes.

Significance of Research

This research will be very important as e-cigarette use is very common in today's society. There are so many options in terms of what to use, as there are different flavors and types of e-cigarettes available. There is also a big push for health through advertising. Weight loss programs and medications, for example, seem to populate major social media platforms and television channels all over the place. People want to know how to be healthier, so providing this information, while not necessarily novel, is pertinent in today's age. While e-cigarettes were initially thought to be healthier, studies have shown that they can be more addictive and just as dangerous as traditional

cigarettes (Rohde et al.; 2020). So, those who are interested in living the healthiest lifestyle would want to stay away from e-cigarettes at all costs.

It will be useful to see whether an individual has a higher likelihood of using an e-cigarette based on their lifestyle choices. Certain lifestyle choices may lead to an individual having a higher chance of using e-cigarettes, so it will be interesting to see which lifestyle choices have the biggest impact on predicting e-cigarette usage.

Novelty

The use of e-cigarettes and their potential harmful impacts is nothing new. People have been asking if they are safe to use ever since they were introduced. What is original, however, is whether a person is more likely to use an e-cigarette than others. Are healthy individuals less likely to use e-cigarettes than those with less healthy lifestyles? Could variables such as BMI, exercise, diet, traditional smoking status, and drug use be used to predict the likelihood that an individual will use an e-cigarette? Are any variables more likely to play a larger role in predicting whether an individual is more likely to use an e-cigarette?

Hypothesis

We hypothesize a predictive model for e-cigarette usage can be built using health and lifestyle data because unhealthy habits frequently co-occur.

Prediction

We predict that those with healthier living habits may be less likely to use e-cigarettes due to their desire to live the healthiest lifestyle.

Data & Methods

The 2024 Behavioral Risk Factor Surveillance System (BRFSS) dataset will be used to answer this question (*2024 BRFSS Survey Data and Documentation*) and was downloaded from the Centers for Disease Control and Prevention (CDC) website. This dataset is a good match for our question because it collects extensive health information annually from U.S. residents in all 50 states, the District of Columbia, Puerto Rico, the U.S. Virgin Islands, and Guam. This voluntary survey covers various health-related topics such as preventative health, current and past diagnoses, mental health, and drug and tobacco use. For our scenario, variables related to e-cigarette use, drug use, tobacco use, alcohol consumption, demographics, physical and mental health, and lifestyle behaviors will all be included. The response variable is _CURECI3, representing whether or not the respondent is a current e-cigarette user. The predictor variables chosen will be used to analyze how demographic information, lifestyle, and health behaviors are associated with e-cigarette use. Our initial plan is to include all of the lifestyle variables and use both exploratory data analysis (EDA) and feature selection to determine which variables should be included in the models.

The original dataset is a SAS Transport File (.XPT), which was read into R and written as a CSV file. From there, the CSV file was loaded into Python using the pandas library. The dataset contains 301 columns and 457,670 rows, with questions and variables in multiple sections. After reviewing the sections for relevant variables, the updated dataset now contains 54 variables and 457,670 rows. Additional variables may

be added or deleted, but this initial selection includes demographic, lifestyle, smoking, drug use, alcohol consumption, physical, and mental health variables. The Data Dictionary is located in Appendix A and includes our initial variable selection. It is subject to change throughout the analysis and the variable descriptions and details have been paraphrased or taken directly from the 2024 BRFSS Codebook.

Limitations

There are multiple limitations with the BRFSS dataset. The first is that participation is voluntary for adults aged 18 and older and individuals are chosen through random-digit dialing (*BRFSS Frequently Asked Questions (FAQs)*). According to the BRFSS 2024 Summary Data Quality Report, the overall response rate was 43.9% (*2024 Summary Data Quality Report*). Less than half of all individuals contacted completed the survey; therefore, a nonresponse bias was introduced, potentially skewing the overall survey results. Additionally, this survey relies on individuals to both answer their phone and be willing to take the entire survey.

A second limitation is the language the survey is offered. According to the BRFSS Overview, it is offered in both English and Spanish (*BRFSS Overview 2024*). This may exclude individuals who speak other languages and can lead to potential bias in the results. Additionally, phone-based surveys may be difficult for some non-English or non-Spanish speaking individuals, leading to additional challenges and the underrepresentation of certain populations.

Since this survey relies on self-reported data, a third limitation is that individuals may inaccurately report physical or mental health conditions, leading to the underestimation or overestimation of certain conditions. This is a common problem with

survey data and one that individuals at Harvard T.H. Chan School of Public Health have seen (*Correcting for BMI Self-Reporting Bias*). Since health information such as height and weight is self-reported, the BMI variable can have self-reporting bias. This can affect other variables in the dataset, such as _PHYS14D or _MENT14D, which represent the number of days the respondent reported poor physical or mental health. These variables provide crucial information for the survey, as they can indicate chronic physical health conditions, such as arthritis, or chronic mental health conditions, such as depression.

Analysis Plan

Initially we will be reducing the huge number of variables from the BRFSS dataset, with a focus on demographics, e-cigarette usage and a list of health based habits such as Heavy Alcohol Usage, Overweight or Obese and Exercise in the past 30 Days.

Then we will begin analyzing any missing data points and determining whether deleting these data points, filling them in with means or medians, or removing the variable entirely is appropriate. With such a large dataset to work with, we will have a number of options here without risking diluting our dataset down too low where the results will be skewed. Our initial plan is to change any blank, “refused”, or “don’t know” data points to NaN’s, and any data listed as “nones” to zero.

Then we will analyze correlation between the variables. This step is crucial because it can lead you down different paths. Some sets of data can be so heavily correlated, regardless of variable selection, that it can determine your model selection. Other data sets will only have a handful of variables that are strongly correlated, which

can lead to stricter variable selection, but a wider possibility of model options. For the purposes of this study, we will need to look at correlations between variables that measure very similar health measures in similar ways. For example the measure exercise in the last 30 days and exercise used to assist with joint or arthritis pain. It's likely this process will lead to us only picking one of these variables.

Then we will be exploring the distribution of data in our remaining variables, to see if these are distributed in a bell curve, or have a spread of outliers. We will also discuss the value of changing certain variables into categories, such as turning BMI into underweight, healthy weight and overweight, to increase interpretability.

In regards to model selection, we are focused on a binary outcome, so we will first be using logistic regression, as this gives us easily interpretable impacts each variable had in predicting likelihood of e-cigarette usage.

As mentioned before, we also risk having too many variables, and ones that are heavily correlated, so this is motivating us to try a Ridge model, which will reduce the impact of correlated variables and offer us stability in coefficients. Lastly, a Random Forest model can handle non-linear relationships and would offer a great comparison to our logistic regression model. For an unsupervised model, we plan to use K-Means-Clustering. This will allow us to see natural forming groups we would not otherwise notice in a supervised model.

In terms of determining if our hypothesis is correct, that healthier people are less likely to smoke e-cigarettes, we can evaluate this using an ROC curve to evaluate the

predictive strength with the goal being the curve being as close as possible to the top left and an area under the curve in our plot above 80%.

Then we can produce a confusion matrix to analyze our performance in three key metrics: precision, recall and F1 score. Precision will be used to make sure we are not over predicting false positives, recall will be used to ensure we are minimizing false negatives, and F1 is used to make sure we are balancing each.

Due to different strengths within our team, we plan to use a combination of R and Python for this project.

Link to Github:

https://github.com/KWilson310/E-Cigarette-and-Healthy-Habits-Study

Dataset

https://www.cdc.gov/brfss/annual_data/2024/files/LLCP2024XPT.zip

Works Cited

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Appendix

Appendix A: Data Dictionary

Variable Name	Data Type	Variable Description	Variable Details	Role in Analysis
_AGEG5YR	Categorical	Age category of respondents	1: Age 18-24 2: Age 25-29 3: Age 30-34 4: Age 35-39 5: Age 40-44 6: Age 45-49 7: Age 50-54 8: Age 55-59 9: Age 60-64 10: Age 65-69 11: Age 70-74 12: Age 75-79 13: Age 80+ 14: Don't know	Predictor variable
ALCDAY4	Numeric	Represents the number of days per week or per month the respondent had at least one drink of an alcoholic beverage	100 101-107: Days per week 201-230: Days in past 30 days 777: Don't know 888: No drinks in the past 30 days 999: Refused Blank	Predictor variable
AVEDRNK4	Numeric	Represents the average number of drinks consumed on a day when the respondent drank in the past 30 days	1-76: Number of drinks 88: No drinks 77: Don't know 99: Refused Blank	Predictor variable
_BMI5	Numeric	The Body Mass Index (BMI) of the respondent	1-9999: 1 or greater Blank	Predictor variable
_BMI5CAT	Categorical	BMI Categories	1: Underweight (_BMI5 < 1850) 2: Normal Weight (1850 <= _BMI5 < 2500) 3: Overweight (2500 <= _BMI5 < 3000) 4: Obese (3000 <= _BMI5 < 9999) Blank	Predictor variable
CHECKUP1	Categorical	How long it has been since the respondent last had a routine checkup	1: Within the past year 2: Within the past two years 3: Within the past five years 4: Five or more years ago 7: Don't know 8: Never 9: Refused Blank	Predictor variable
CHILDREN	Numeric	Number of children under 18 years old living in the respondent's household	0 1-87: Number of children 88: None 99: Refused Blank	Predictor variable
_CURECI3	Categorical	Represents whether or not the respondent is a current e-cigarette user	1: Not a current e-cigarette user 2: Is a current e-cigarette user 9: Don't know	Response/Outcome variable
CVDCRHD4	Categorical	Whether the respondent has ever been told they have angina or coronary heart disease	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
_DENVST3	Categorical	Whether the respondent has visited a dentist in the past year	1: Yes 2: No 9: Don't know	Predictor variable
DIABETE4	Categorical	Whether the respondent has ever been told they have diabetes	1: Yes 2: Yes, but only female during pregnancy 3: No 4: No (pre-diabetes or borderline) 7: Don't know 9: Refused Blank	Predictor variable
ECIGNOW3	Categorical	Represents the current e-cigarette usage status of the respondent	1: Never used e-cigarettes 2: Use e-cigarettes every day 3: Use e-cigarettes some days 4: Previous user, but not anymore 7: Don't know 9: Refused Blank	Predictor variable
EDUCA	Categorical	Respondent's highest level of education	1: Never attended school or only kindergarten 2: Grades 1-8 (elementary) 3: Grades 9-11 (some high school) 4: Grade 12 or GED (high school graduate) 5: College 1-3 years (some college or technical school) 6: College 4 years or more (college graduate) 9: Refused Blank	Predictor variable
EMTSUPRT	Categorical	How often the respondent gets the social and emotional support that they need	1: Always 2: Usually 3: Sometimes 4: Rarely 5: Never 7: Don't know 9: Refused Blank	Predictor variable

EXERANY2	Categorical	Whether or not the respondent has exercised outside of work in the past month	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
GENHLTH	Categorical	How the respondent thinks their general health is	1: Excellent 2: Very good 3: Good 4: Fair 5: Poor 7: Don't Know/Not Sure 9: Refused Blank	Predictor variable
_HLTHPL2	Categorical	Calculated variable that represents whether or not the respondent has health insurance	1: Have some form of insurance 2: Does not have health insurance 9: Don't know	Predictor variable
INCOME3	Categorical	Annual household income of the respondent	1: Less than $10,000 2: Less than $15,000 3: Less than $20,000 4: Less than $25,000 5: Less than $35,000 6: Less than $50,000 7: Less than $75,000 8: Less than $100,000 9: Less than $150,000 10: Less than $200,000 11: 200,000+ 77: Don't know 99: Refused Blank	Predictor variable
LASTSMK2	Categorical	How long it has been since the respondent last smoked a cigarette	1: Within the past month 2: Within the past three months 3: Within the past six months 4: Within the past year 5: Within the past five years 6: Within the past ten years 7: Ten years or more 8: Never smoked regularly 77: Don't know 99: Refused Blank	Predictor variable
_LCSAGE	Categorical	Calculated variable that represents lung cancer screening age groups	1: 18-49 2: 50-80 3: 81+	Predictor variable
_LCSCTSN	Categorical	Whether or not the respondent has had a chest CT scan for lung cancer	1: Had a chest CT scan for lung cancer within 1 year 2: Had a chest CT scan for lung cancer over 1 year ago 3: Had a chest CT scan for other reason 4: Never had a chest CT scan Blank	Predictor variable
LCSLAST_	Numeric	Calculated variable representing the age the respondent was when they last smoked cigarettes regularly	1-80: Age in years 81: Age in years (81+) Blank	Predictor variable
LCSNUMC_	Numeric	When smoking regularly, how many cigarettes the respondent smokes per day	1-300: Number of cigarettes Blank	Predictor variable
_LCSYSMK	Numeric	The number of years the respondent smoked cigarettes	0-100: Number of years Blank	Predictor variable
_LCSSMKG	Categorical	LCS (Lung Cancer Screening) Smoking Group status	1: Current smoker (20+ packs per year) 2: Former smoker (20+ packs per year), quit < 15 years 3: Current smoker (less than 20 packs per year) 4: Former smoker (20+ packs per year), quit >= 15 years 5: Former smoker (< 20 packs per year) 6: Never smoked Blank	Predictor variable
_LCSYQTS	Numeric	Number of years since the respondent quit smoking	0-100: Number of years since they last smoked cigarettes Blank Hidden	Predictor variable
LSATISFY	Categorical	Level of satisfaction in the respondent's life	1: Very satisfied 2: Satisfied 3: Dissatisfied 4: Very dissatisfied 7: Don't know 9: Refused Blank	Predictor variable
MARIJAN1	Numeric	How many days the respondent has used marijuana or cannabis in the past 30 days	1-30: Number of Days 88: None 77: Not Sure 99: Refused Blank	Predictor variable
MARITAL	Categorical	Marital status of the respondent	1: Married 2: Divorced 3: Widowed 4: Separated 5: Never married 6: A member of an unmarried couple 9: Refused Blank	Predictor variable

MEDCOST1	Categorical	Whether or not there was a time in the past 12 months the respondent needed to see a doctor but couldn't because of the cost	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
MENTCIGS	Categorical	Whether the respondent usually uses menthol cigarettes when currently smoking	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
MENTECIG	Categorical	Whether the respondent usually uses menthol e-cigarettes when currently using e-cigarettes	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
_MENT14D	Categorical	Number of days the respondent had poor mental health	1: 0 days 2: 1-13 days 3: 14+ days 9: don't know/refused/missing	Predictor variable
MENTHLTH	Numeric	Indicates the number of days in the past 30 days the respondent's mental health was not good	1-30: Number of Days 88: None 77: Don't Know/Not Sure 99: Refused Blank	Predictor variable
_PACKDAY	Numeric	Calculated variable that represents the number of packs of cigarettes the respondent smokes per day	0-100: Number of packs smoked per day Blank	Predictor variable
_PACKYRS	Numeric	Calculated variable that represents the number of years the individual has smoked their reported number of packs per day	0-999: Number of years Blank	Predictor variable
_PHYS14D	Categorical	Number of days the respondent had poor physical health	1: 0 days 2: 1-13 days 3: 14+ days 9: Don't know	Predictor variable
PHYSHLTH	Numeric	Indicates the number of days in the past 30 days the respondent's physical health was not good	1-30: Number of Days 88: None 77: Don't know 99: Refused Blank	Predictor variable
POORHLTH	Numeric	Indicates number of days in the past 30 days the respondent had poor physical or mental health that kept them from doing their regular activities	1-30: Number of Days 88: None 77: Don't know 99: Refused Blank	Predictor variable
PREDIAB2	Categorical	Whether the respondent has ever been told they have pre-diabetes	1: Yes 2: Yes, during pregnancy 3: No 7: Don't know 9: Refused Blank	Predictor variable
_RACE	Categorical	Race of the respondent	1: White only, Non-Hispanic 2: Black only, Non-Hispanic 3: American Indian or Alaskan Native only, Non-Hispanic 4: Asian only, Non-Hispanic 5: Native Hawaiian or other Pacific Islander only, Non-Hispanic 6: Other race only, Non-Hispanic 7: Multiracial, Non-Hispanic 8: Hispanic 9: Don't know	Predictor variable
_RFBMI5	Categorical	Respondents with a BMI > 25, classifying them as overweight or obese	1: No 2: Yes 9: Blank	Predictor variable
_RFHLTH	Categorical	A calculated variable representing what respondents think of their health	1: Good or Better Health (GENHLTH = 1 or 2 or 3) 2: Fair or Poor Health (GENHLTH = 4 or 5) 9: Don't know/Not sure Or Refused/Missing (GENHLTH = 7 or 9 or Missing)	Predictor variable
_RFSMOK3	Categorical	Current smoking status of respondents	1: No 2: Yes 9: Don't know	Predictor variable
SEXVAR	Categorical	Sex of the respondent	1: Male 2: Female	Predictor variable
SDLONELY	Categorical	How often the respondent feels lonely	1: Always 2: Usually 3: Sometimes 4: Rarely 5: Never 7: Don't know 9: Refused Blank	Predictor variable
SMOKE100	Categorical	Whether or not the respondent has smoked at least 100 cigarettes in their lifetime	1: Yes 2: No 7: Don't Know 9: Refused Blank	Predictor variable

SMOKDAY2	Categorical	Whether or not the respondent currently smokes cigarettes every day, some days, or not at all	1: Every day 2: Some days 3: Not at all 7: Don't Know 9: Refused Blank	Predictor variable
_SMOKER3	Categorical	Smoker status	1: Current smoker (smokes every day) 2: Current smoker (smokes some days) 3: Former smoker 4: Never smoked 9: Don't know Blank	Predictor variable
SSBSUGR2	Numeric	The number of times the respondent drank soda or pop that contained sugar in the past 30 days	101-199: Times per day 201-299: Times per week 301-399: Times per month 777: Don't know 888: Never 999: Refused Blank	Predictor variable
SSBFRUT3	Numeric	The number of times the respondent drank fruit juice, sweet tea, sports, or energy drinks that contained sugar in the past 30 days	101-199: Times per day 201-299: Times per week 301-399: Times per month 777: Don't know 888: Never 999: Refused Blank	Predictor variable
STOPSMK2	Categorical	Whether the respondent has stopped smoking for one day or longer when trying to quit in the past 12 months	1: Yes 2: No 7: Don't know 9: Refused Blank	Predictor variable
_TOTINDA	Categorical	A calculated variable that represents whether or not the respondent has performed physical activity or exercised in the past 30 days outside of their job	1: Had physical activity or exercise (EXERANY2 = 1) 2: No physical activity or exercise in last 30 days (EXERANY2 = 2) 9: Don't know/Refused/Missing (EXERANY2 = 7 or 9 or Missing)	Predictor variable
USENOW3	Categorical	How often the respondent currently uses chewing tobacco, snuff, or snus	1: Every day 2: Some days 3: Not at all 7: Don't know 9: Refused Blank	Predictor variable